

"From the Nature for the Nature": An Eco-Friendly Antifouling Coating Consisting of Poly(lactic acid)-Based Polyurethane and Natural Antifoulant

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Cite This: *ACS Sustainable Chem. Eng.* 2020, 8, 1671–1678



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ABSTRACT: "From the nature, for the nature" is an instructive strategy to develop environment-friendly antifouling coatings. In this work, bio-sourced poly(lactic acid)-based polyurethane with hydrolyzable triisopropylsilyl acrylate (TSA) side groups has been prepared via thiol–ene reaction and polyaddition. Such a polymer exhibits high adhesion strength (~ 2.0 MPa) and a controlled degradation rate tuned by varying its soft segment and TSA content. An environment-friendly coating is prepared with the polymer and an eco-friendly antifoulant (butenolide) derived from marine bacteria. Our study shows that butenolide can be released from the coating in continuous manner with a controlled rate as the polymer degrades in seawater. Digital holographic microscopy tracking analysis demonstrates that the coating can effectively inhibit the adhesion of marine bacteria *Pseudomonas* sp. Marine field test shows that such coating has excellent antifouling ability for more than 3 months.

KEYWORDS: Bio-based polymer, poly(lactic acid), natural compound, butenolide, controlled release, antifouling coating



INTRODUCTION

Marine biofouling causes serious economic loss and safety problems for maritime industries.^{1–3} The incentive and demand for preventing marine biofouling have always been significant and high. With the enhancement of people's environmental awareness and the increasingly strict regulations, eco-friendly antifouling systems have become the focus of current research. To this end, fouling release coatings,^{4–8} self-generated hydrogels,^{9–13} zwitterionic polymers,^{14–19} amphiphilic polymers,^{20–24} and biomimetic antifouling materials^{25,26} were prepared. In particular, degradable polymers with dynamic surface can significantly inhibit marine biofouling from adhering.^{27,28} In addition, their dynamic surface enables the continuous release of antifoulant. As such, degradable polymers are promising in combating biofouling.

Using natural-sourced raw materials to prepare degradable polymers is becoming increasingly attractive.^{29,30} Poly(lactic acid) (PLA), usually derived from crops such as corn, wheat, and sugarcane, is one of the most widely used bio-based and biodegradable polyesters. It is a green choice to replace petroleum-based polymers because PLA degrades quickly and completely into nontoxic carbon dioxide and water, which finally returns to the environment.³¹ Besides, because of its excellent biodegradability and biocompatibility, PLA is approved by the US Food and Drug Administration (FDA) for biomedical applications^{32,33} and finds many other

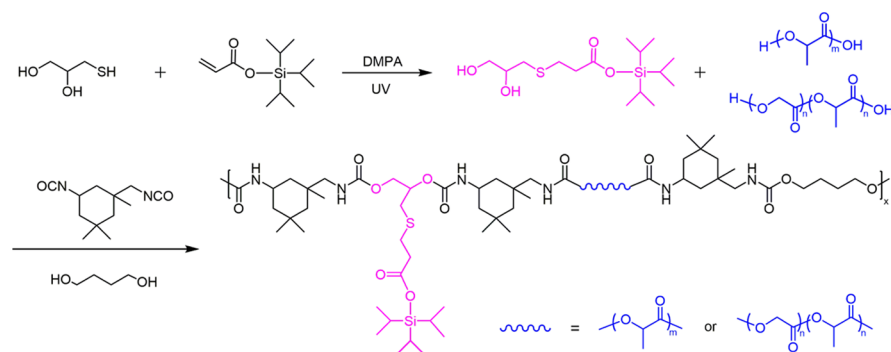
applications in food packaging, composite membrane, coatings, and so on.^{34–36} Owing to its unique bio-based and eco-friendly nature, it makes significance to the development of marine antifouling coatings. However, high molar mass PLA has slow degradation and poor adhesion strength because of the high degree of crystallinity, and it is prone to crack and detach from substrate, which limits its application in marine antifouling coatings.²⁷ Thus, chemical modification on PLA such as using low molar mass PLA segments to synthesize polyurethane can make it a suitable polymer binder for antifouling coatings. Meanwhile, low molar mass PLA can ensure the high reactivity of terminal hydroxyl group with NCO in the synthesis of polyurethanes.³⁷ Furthermore, the degradation and crystallinity of PLA can be easily adjusted by copolymerization of other cyclic monomer such as glycolide (GA).³⁸

In addition to using bio-based polymer as polymer binder, incorporating environmentally benign antifouling compounds can further minimize the environmental impact from the current commercial antifouling coatings while enhancing the antifouling efficiency.³⁹ Natural product antifoulants (NPAs)

Received: November 19, 2019

Revised: December 13, 2019

Published: December 31, 2019

Scheme 1. Synthetic Route of Y-PU-*x*Table 1. Characterization Data of Y-PU-*x*

sample	Y/IPDI/BDO/TSA ^a	TSA content ^b	<i>M_n</i> (g/mol) ^c	<i>D_M</i> ^c	<i>T_g</i> (°C) ^d	pencil hardness
PLA-PU-0	50.0/37.2/12.8/0	0	17,000	1.6	60.5	4H
PLA-PU-4	50.0/35.1/10.9/4.0	3.8	16,000	1.7	56.5	4H
PLA-PU-10	50.0/32.0/8.0/10.0	10.3	15,000	1.5	48.6	4H
PLA-PU-16	50.0/28.9/5.1/16.0	16.2	19,000	1.7	42.8	3H
PLGA-PU-16	50.0/28.9/5.1/16.0	15.9	18,000	1.8	39.1	2H

^aFeed weight ratio. ^bDetermined by ¹H NMR. ^cDetermined by GPC. ^dDetermined by DSC.

extracted from natural sources are an ideal choice for antifouling coatings because they exhibit excellent eco-friendliness. Until now, butenolide, terpenoids, camptothecin, and phenolic compounds from marine bacteria, algae, and terrestrial plants have been reported to effectively inhibit the attachment of diverse fouling organisms.^{40–43} Moreover, NPA-based antifouling coatings have also been explored in the past few years. For instance, coatings containing camptothecin derived from terrestrial plants have been found to be effective in laboratory bioassays and field trials.⁴² However, the complex structure and high cost of camptothecin make it challenging for commercial marine antifouling coating application. Coatings consisting of tannins and nonreactive polymers such as vinyl resin or chlorinated rubber have also been reported.⁴⁴ Although the content of tannins in coatings is higher than 27 wt %, the release rate of tannins dramatically decreases with time because of the limited surface renewal ability of the polymer binder. Biodegradable coatings based on hemibastadin derivatives were developed to reduce the biofilm development.⁴⁵ Yet, similar obstacles such as complex structure and poor yield of hemibastadin derivatives limit their application. Actually, the use of NPAs into antifouling coatings is rather slow because they face several practical challenges such as cost-effective mass production, compatibility with polymer binder, and controlled release from coatings.

Herein, we reported a novel strategy for anti-biofouling “from the nature, for the nature” and developed an eco-friendly coating based on butenolide and PLA polyurethane. Butenolide derived from marine bacteria has been proved to be a promising natural antifoulant because it has high antifouling activity, low toxicity, and rapid degradation. More importantly, it has simple structure and can be synthesized in large scale economically.^{46,47} On the other hand, bio-based and biodegradable polyurethane was used as the polymer binder to carry butenolide and control its release. Polyurethane contains PLA or poly(lactide-co-glycolide) as soft segments and triisopropylsilyl acrylate (TSA) pendant groups. With degradable main chains and hydrolyzable side groups, the polymer

can regulate the surface renewal rate, which is expected to optimize butenolide release behavior and prolong the service life. We have studied the adhesion strength and hydrolytic degradation of the polyurethane with different types of soft segments and content of TSA, the release rate of butenolide, and antifouling performance of the coating through laboratory and marine field test. The highlight of this system is that both polymer binder and antifoulant are from nature and for nature. Our goal is to develop a green but high-performance antifouling system.

EXPERIMENTAL SECTION

Materials. Poly(L-lactic acid)diol (PLA, *M_w* = 2000 g/mol, amorphous with a *T_g* value of 0 °C), poly(L-lactic-co-glycolic acid)diol (PLGA, molar ratio of LA/GA: 50/50, *M_w* = 2000 g/mol, amorphous with a *T_g* value of −3.5 °C) from Daigang Biomaterial (Jinan) and 1,4-butanediol (BDO) from Aladdin were dried under reduced pressure at 110 °C for 2 h. TSA, thioglycerol, isophorone diisocyanate (IPDI), 2,2-dimethylol propionic acid, and dibutyltin dilaurate (DBTDL) from Aladdin were used as received. Tetrahydrofuran (THF) was refluxed over CaH₂ and distilled prior to use. 5-Octylfuran-2(5*H*)-one (butenolide) with a purity of >99% was purchased from ChemPartner (Shanghai). Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2013).

Synthesis of PLA-Based Polyurethane with TSA Pendant Groups. PLA-based polyurethanes with TSA pendant groups were synthesized by combining thiol–ene reaction and polyaddition (Scheme 1). The synthetic procedure of TSA with terminal dihydroxy was reported elsewhere.⁴⁸ Firstly, 5.00 g (2.5 mmol) of PLA diol and 2.89 g (13.0 mmol) of IPDI were stirred under N₂ atmosphere with THF as solvent at 70 °C for 1 h. Then, 1.60 g (4.8 mmol) of TSA and 0.02 g of DBTDL were added, and the reaction was kept at 80 °C for 1 h. Finally, 0.51 g (5.7 mmol) of BDO was added and the mixture was allowed to react for 3 h. The yielded polymer was precipitated in excess hexane twice, filtered, and dried under vacuum at 60 °C for 24 h. ¹H NMR (ppm): 5.16 (HNCOOCH(CH₃)COO), 4.80 (HNCOOCH₂COO), 1.57 (HNCOOCH(CH₃)COO), 4.07 (OCH₂CH₂CH₂CH₂O), 1.69 (OCH₂CH₂CH₂CH₂O), 3.79 (CH₂CH(CH₂)NHCOO), 2.91 (OOCNHCH₂C(CH₂)₂CH₃), 1.28 (SiCH(CH₃)₂), and 1.05 (SiCH(CH₃)₂). Fourier transform infrared (FTIR): 3400 cm^{−1} (NH), 3325 cm^{−1} (hydrogen-bonded NH), 1750

cm^{-1} ($\text{C}=\text{O}$ in TSA, PLA, and free urethane), 1705 cm^{-1} ($\text{C}=\text{O}$ in hydrogen-bonded urethane), and 883 cm^{-1} ($\text{Si}-\text{C}$). The details can be found in Figures S1 and S2. For convenience, the polyurethane with TSA pendant groups is designed as Y-PU- x , where Y represents PLA or PLGA, and x is the weight percentage of TSA determined by ^1H NMR. The characterization data are summarized in Table 1.

Proton Nuclear Magnetic Resonance Spectroscopy (^1H NMR). Y-PU- x was first dissolved in CDCl_3 with tetramethylsilane and then subjected to a Bruker spectrometer (AV600 NMR).

Fourier Transform Infrared Spectroscopy. The spectra of Y-PU- x were recorded on a Bruker spectrometer (VECTOR-22 IR).

Gel Permeation Chromatography. The molar mass (M_n) and molar mass dispersity (D_M) were determined on a Waters 1515 with polystyrenes standard and summarized in Table 1. The tests were carried out at $50\text{ }^\circ\text{C}$ under 1.0 mL/min eluent of dimethylformamide and LiBr.

Differential Scanning Calorimetry. Differential scanning calorimetry (DSC) measurements were performed on a differential scanning calorimeter (Netzsch DSC 204F1) under a 50 mL/min nitrogen flow. The sample was quickly heated to $130\text{ }^\circ\text{C}$ and kept for 5 min to remove thermal history. Then, it was cooled down to $-50\text{ }^\circ\text{C}$ at a rate of $10\text{ }^\circ\text{C/min}$ and equilibrated for 5 min . Subsequently, it was reheated to $130\text{ }^\circ\text{C}$ at the same rate. The glass-transition temperature (T_g) was obtained from the midpoint of the endothermic shift during the second heating scan. All of the soft segments and polyurethanes are amorphous (Figures S4 and S5). The data were summarized in Table 1. Clearly, T_g of Y-PU- x decreases with the TSA content. This is because TSA in hard segments restrict the formation of hydrogen bond between urethane groups reflected by the shift of wavenumber for NH from 3325 to 3400 cm^{-1} (Figure S2), which can improve the motion of main chains. In other words, TSA could increase the flexibility of the copolymer main chains.

Preparation of Polymer Films. Each $(300 \pm 10)\text{ }\mu\text{m}$ polymer film was prepared by a solution casting method. A solution ($30\%\text{ w/v}$) consisting of polyurethane and THF was coated on a glass fiber-reinforced epoxy resin panel, and they were dried at room temperature until the solvent evaporated completely. The coatings with a thickness of $\sim 300\text{ }\mu\text{m}$ are reproducible by varying the amount of polyurethane solution based on the area of substrate. The yielded films were used for adhesion test, water contact angle (WCA) measurement, hydrolytic degradation and water adsorption (WA) test, butenolide release rate measurement, and marine field test.

Surface Hardness Test. The surface hardness of the coatings was measured by pencil test according to ISO 15184, which is widely used in antifouling coatings.^{49,50}

Surface Morphology Measurements. The surface morphology of Y-PU- x and their coatings with butenolide was measured on a surface profile measurement system (BMT Expert 3D). For each sample, the surface morphology in an area of $1 \times 1\text{ mm}^2$ was observed by area scan mode.

Adhesion Test. A pull-off adhesion tester (PosiTest AT-A Automatic) was used to measure the adhesion strength of Y-PU- x according to ASTM D4541. The value of each sample was calculated by averaging five data points.

WCA Measurement. The WCA of Y-PU- x was measured on a Theta Auto 113 (KSV NIMA, Biolin Scientific) by dropping $3\text{ }\mu\text{L}$ of deionized water using a sessile method. Before the measurements, the polymer films were taken out from a tank containing ASW after a predetermined interval, rinsed with deionized water, and dried with nitrogen. The value of each sample was calculated by averaging five data points.

Hydrolytic Degradation and WA Test. The hydrolytic degradation test was conducted in ASW at $25\text{ }^\circ\text{C}$. The polyurethane solution was coated on an epoxy resin panel with an area of 4 cm^2 . The weight (W_0) of each dried film together with its panel was weighted before immersion in ASW. After a given period of time, the sample was taken out, rinsed with deionized water, and dried with nitrogen. The weight of the wet sample was recorded as $W_{\text{t,wet}}$. Then, the sample was dried under vacuum until a constant weight ($W_{\text{t,dry}}$). The mass loss was calculated from $(W_0 - W_{\text{t,dry}})/4$, while the WA was

calculated from $(W_{\text{t,wet}} - W_{\text{t,dry}})/4$. Each data point of Y-PU- x was averaged from three duplicate samples.

Butenolide Release Rate Measurement. Butenolide release rate under static condition was measured by high-performance liquid chromatography (Agilent 1100) equipped with an Eclipse Plus C18 column ($2.1 \times 150\text{ mm}$, $5\text{ }\mu\text{m}$). The procedure was reported in detail elsewhere.³⁹ The release rate of butenolide from the coatings (R , $\mu\text{g}\cdot\text{cm}^{-2}\cdot\text{d}^{-1}$) was calculated from eq 1

$$R = C \times 15 / (D \times A) \quad (1)$$

where C is the butenolide content in methanol (μg) calculated from butenolide peak area and established standard curves, 15 is the times of ASW volume used for immersion and that used for extraction, D is the immersion time in the individual measuring container (1 day in this study), and A is the area of coated films (18.75 cm^2 in this study).

Digital Holographic Microscopy. The films on coverslips ($22 \times 22\text{ mm}^2$) were prepared by spin-coating method with polymer/butenolide coatings in THF ($15\%\text{ w/v}$). The coated coverslips were subsequently dried at $60\text{ }^\circ\text{C}$ for 12 h . *Pseudomonas* sp. NOV776 used in this study is a kind of model marine bacteria for antifouling assay. Details for the bacterial strains, culture conditions, and medium components were reported before.⁵¹ Bacterial adhesion and density distribution were observed by a home-made digital holographic microscopy (DHM) with a resolution of approximately 200 nm , which is much larger than the roughness of coating surfaces ($\sim 30\text{ nm}$). Thus, the details of surface morphology are invisible, and we can easily distinguish bacteria from the background. The working principle of the DHM was detailed elsewhere.⁵² Briefly, bacterial adhesion was defined as the number of bacteria adhered onto the background images (N_b). The density distribution of bacteria with active motion in dependence of their instantaneous 3D velocities and distances from the surface (z) was plotted as heat maps.

Electrochemical Impedance Spectroscopy Measurement.

The polymer solution was coated on one surface of carbon steel panel via a solution casting method. They were then dried at room temperature for 7 days to prepare polymer films with a dry thickness of $(300 \pm 10)\text{ }\mu\text{m}$. To prevent corrosion occurred on the other surface of carbon steel panel, paraffin as a sealant was applied prior to immersion. All the samples were immersed in ASW for 24 h . The barrier property was tested on a Zahner/Zennium potentiostat-galvanostat (Zahner, Germany) equipped with a common three-electrode cell arrangement. The coated carbon steel panel, Pt mesh, and saturated calomel electrode served as working, counter, and reference electrode, respectively. The frequency range of the measurements was from 100 kHz to 100 mHz with an amplitude perturbation of 100 mV .

Marine Field Test. The field test was conducted according to ASTM D6990-05 (2011) method.^{53,54} The test lasted from January to April 2019 at Yung Shue O, Hong Kong ($114^\circ 21'\text{E}$, $22^\circ 24'\text{N}$), where the fouling pressure is high throughout the year. The polyvinyl chloride (PVC) panels coated with the samples ($90\text{ wt } \%$ of polymer with $10\text{ wt } \%$ of butenolide) were hung in seawater ($\sim 0.5\text{ m}$). After the interval of every 30 days , the panels were taken out, gently washed with seawater from the immersion site to remove silt and unattached forms, photographed, and immediately placed back to continue the test. The quantitative data of the area covered by biofouling were determined by ImageJ.

RESULTS AND DISCUSSION

Considering that the marine environment is complicated and service condition is harsh, we first measured the adhesion strength of Y-PU- x to the substrate (Figure 1). PLA-PU-0 has an adhesion strength of 1.71 MPa . For PLA-PU-4, PLA-PU-10, and PLA-PU-16, the values are 1.83 , 1.89 , and 2.07 , respectively, indicating that the adhesion strength increases with TSA content even though the change is slight. As indicated by the T_g values in Table 1, the incorporation of TSA can improve the flexibility of polymer chains, thus increasing

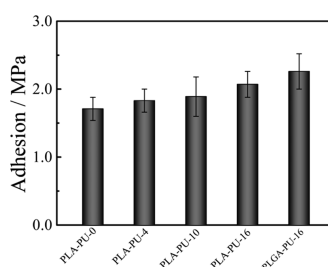


Figure 1. Adhesion strength of Y-PU-*x* on epoxy substrate.

the contact of the esters in main chains and pendant groups to the substrate forming more hydrogen bonds between them.⁵⁵ On the other hand, the higher content of TSA can increase the polar interaction between the polymer and epoxy substrate.⁵⁶ As a result, the adhesion strength is improved by TSA content. Note that PLGA-PU-16 has a higher adhesion (2.26 MPa) than PLA-PU-16. Similar behavior can be observed between PLGA-PU-0 with a T_g of 49.2 °C and PLA-PU-0 with a T_g of 60.5 °C (Figure S7). This is because PLGA segments with less content of CH_3 groups are more flexible than PLA segments; thus, more hydrogen bonds formed between the esters in main chains and substrate. Anyhow, each polymer has an adhesion strength of about 2.0 MPa, which can meet the requirement for the marine applications. Note that strain–stress curves can give the information about the mechanical properties of coatings. We tried to conduct it by tensile test. However, all the samples cracked during demolding, so we cannot obtain dumbbell-shaped specimens either by film-forming in polytetrafluoroethylene mold or compression molding. Actually, strain–stress curves generally reflect the mechanical performance of bulk materials, while adhesion strength is a basic but critical parameter for coatings. We also conducted the surface hardness and surface morphology tests to evaluate the properties of Y-PU-*x* as coatings (Table 1 and Figure S6). The surface hardness of Y-PU-*x* slightly decreases as the content of TSA increases. Anyhow, all the samples have high hardness values above 2H, which can prevent mechanical damage on the coating surface during the service life. Besides, all Y-PU-*x* coatings have smooth surfaces with a roughness of ~30 nm. The barrier property of the polyurethanes was also evaluated by electrochemical impedance spectroscopy measurement (Figure S8). Before immersion, all the coatings have a high impedance value of over $10^9 \Omega$, indicating their good barrier property. After immersion for 24 h, the impedance values in low-frequency region and the radius of capacitive arc decrease, and the changed values decrease with increasing TSA content. Moreover, PLGA-PU-16 has the lowest values. The fact indicates that the barrier property of the coatings decreases with increasing TSA content after immersion, which is probably because of the increasing hydrophilicity and the higher WA. Anyhow, all the coatings have a certain degree of barrier property. Actually, there are limited numbers of reports on the barrier properties of antifouling coatings. In industrial applications, it is always used together with anticorrosive coating, which can provide barrier properties.

Figure 2 shows the time dependence of WCA of Y-PU-*x* surfaces after immersion in ASW. The initial WCA of PLA-PU-0 is 76°. However, WCA increases with TSA content. The values for PLA-PU-4, PLA-PU-10, and PLA-PU-16 are 86, 87, and 88°, respectively. This is because more TSA groups that are hydrophobic covered on the surface as its content

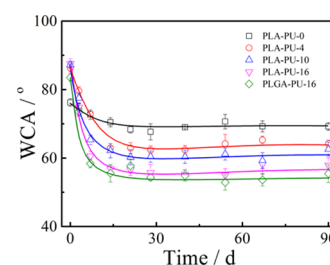


Figure 2. Time dependence of WCA of Y-PU-*x* in ASW at 25 °C.

increases.⁵⁷ The WCA of Y-PU-*x* sharply decreases after 3 days and gradually reaches equilibrium after 14 days. For PLA-PU-0, the WCA decreases to 69° because of the hydrolysis in PLA and the reconstruction of hydrophilic urethane groups on the surface.^{4,58} For PLA-PU-4, PLA-PU-10, and PLA-PU-16, the equilibrated WCA are 64, 62, and 58° respectively. Clearly, the surface becomes more hydrophilic. This is because the hydrophilic carboxyl groups on the surface from the hydrolysis of triisopropylsilyl ester groups increase with TSA content.^{59,60} On the other hand, PLGA-PU-16 has a lower equilibrated WCA than PLA-PU-16. To see any effect of CH_3 groups in PLA on the WCA, we also measured the WCA of PLGA-PU-0 (Figure S9). Similarly, PLGA-PU-0 has a lower WCA than PLA-PU-0. Note that the content of main-chain ester in PLA and PLGA is the same, but PLGA has less hydrophobic CH_3 groups than PLA. The decrease of WCA should be attributed to the synergistic effects of higher hydrolysis rate of PLGA (Figure 3) and the plasticizing effect of the adsorption of

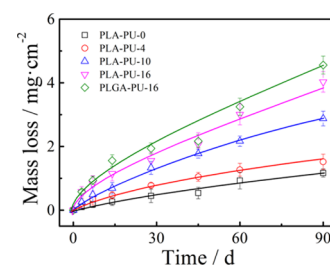


Figure 3. Time dependence of mass loss of Y-PU-*x* in ASW at 25 °C.

water.⁶¹ This is because the former can generate more hydrophilic carboxyl groups, while the latter increases the diffusion of water, which facilitates the attack to ester bond.

Figure 3 shows the time dependence of mass loss of Y-PU-*x* in ASW. In general, each sample has a near-linear hydrolytic degradation rate within 90 days, indicating a constant erosion rate in seawater. PLA-PU-0 without TSA pendant groups shows a mass loss of 1.17 $\text{mg}\cdot\text{cm}^{-2}$ after 90 days because of the cleavage of ester bond in PLA.⁶² For PLA-PU-*x* with the same PLA content, the mass loss increases with TSA content. This is understandable because the hydrolyzable TSA facilitates polymer erosion. PLGA-PU-16 has a slightly higher mass loss than PLA-PU-16. To eliminate the effect of TSA, we also conducted mass loss measurement of PLGA-PU-0 (Figure S10). The result is similar to that of PLA-PU-16 and PLGA-PU-16. PLGA-PU-0 has a higher mass loss than PLA-PU-0. This is because the presence of the bulky methyl groups and the lower flexibility in PLA decrease water diffusion and inhibit the subsequent attack to ester bond. In other words, we can regulate the surface renewal rate of polymer binder by varying

both main chains and side groups to apply under different conditions.

Figure 4 shows the time dependence of WA of Y-PU-*x* in ASW. PLA-PU-0 has a low WA of $\sim 1.6 \text{ mg}\cdot\text{cm}^{-2}$ after 90 days.

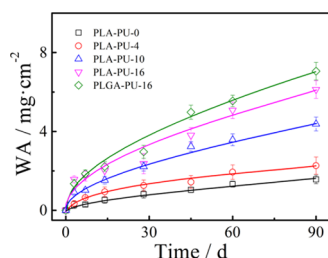


Figure 4. Time dependence of WA of Y-PU-*x* in ASW at 25 °C.

For PLA-PU-*x*, the WA increases with TSA content. This is because more TSA generates more hydrophilic $-\text{COOH}$ after hydrolysis. For PLGA-PU-16 and PLA-PU-16 with the same ester density but different CH_3 contents, the former has a higher WA in that the hydrophilicity increases as the CH_3 content decreases. Nevertheless, heavy swelling or detachment from substrate was not observed for any sample, indicating that the polymer is suitable for long-term immersion in seawater.

The controlled release of antifoulant is critical for prolonging the duration and improving the antifouling ability of an antifouling system. Figure 5 shows the time dependence of 10

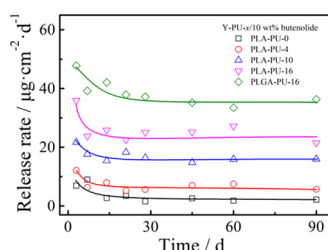


Figure 5. Time dependence of release rate of 10 wt % butenolide from Y-PU-*x* in ASW at 25 °C.

wt % butenolide release rate from Y-PU-*x* in ASW at 25 °C. For PLA-based coatings, the release rates are relatively high at the early stage but gradually decrease and level off after 14 days. The equilibrated release rate is $\sim 3, 7, 16$, and $23 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{d}^{-1}$, respectively, indicating that the release rate increases with TSA content. Thus, we can regulate butenolide release by varying the content of side groups. In comparison, PLGA-based coating has a higher rate ($36 \mu\text{g}\cdot\text{cm}^{-2}\cdot\text{d}^{-1}$) than PLA-based coatings. The same trend can be observed in degradation rate. Clearly, the release rate of butenolide can also be regulated by changing the types of soft segments besides TSA content.

To gain insight into the antifouling mechanism of the Y-PU-*x*/butenolide coating, we tracked the motion behavior of *Pseudomonas* sp. near the coating surface (Figure 6). The adhered number (N_b) of *Pseudomonas* sp. on PLA-PU-0 without butenolide is $4.9 \times 10^5\cdot\text{cm}^{-2}$. The value is much higher than the coatings with butenolide, indicating that the presence of antifoulant can inhibit the adhesion of marine bacteria. Particularly, PLGA-PU-16 has a significantly lower N_b than PLA-PU-0. As shown in Figures 3 and 5, the former has a higher hydrolytic degradation rate and release rate of antifoulant after immersing in ASW. Accordingly, a higher

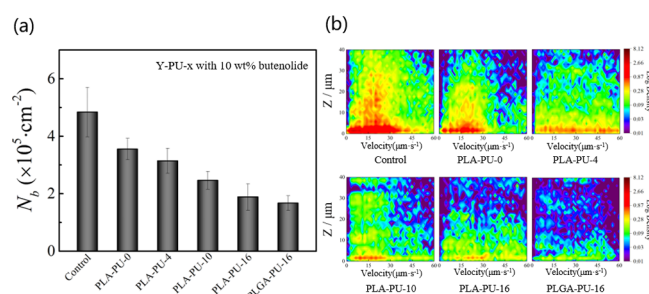


Figure 6. (a) The number of *Pseudomonas* sp. adhered onto the coating surfaces consisting of Y-PU-*x* and 10 wt % butenolide. (b) Density distribution (heat maps) of instantaneous 3D velocities and distances from the coating surfaces consisting of Y-PU-*x* and 10 wt % butenolide for *Pseudomonas* sp. Color bars represent the relative density of data points with a logarithmic scale. The control sample is PLA-PU-0 without butenolide.

release rate leads to a higher antifoulant concentration near the coating surface and better antifouling ability. The density distribution of *Pseudomonas* sp. near the coating surfaces shows a trend similar to N_b . Fewer bacteria distribute near PLGA-PU-16 than other samples, indicating that the bacteria to approach the antifoulant-enriched surface are more reluctant. These results clearly show that the release rate of antifoulant is critical for antifouling property. Theoretically, the coatings with a linear mass loss rate can be always active after immersion in seawater till the polymer binder degrades completely.

The long-term antifouling performance of Y-PU-*x*/butenolide coatings (90 wt % polymer with 10 wt % antifoulant) was evaluated in South China Sea. Figure 7 shows the images of

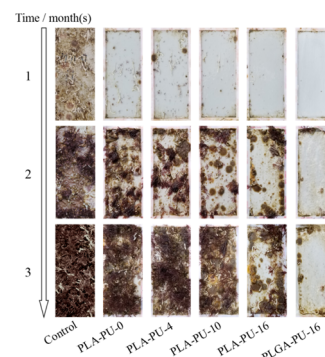


Figure 7. Images of coatings consisting of Y-PU-*x* and 10 wt % butenolide after immersion in natural seawater for 3 months (Yung Shue O, Hong Kong).

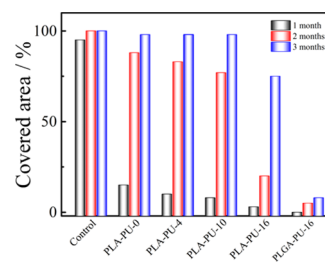


Figure 8. Quantitative analysis of biofouling covered area of Y-PU-*x* with 10 wt % butenolide coatings in marine field test.

coating surfaces after immersion for 3 months, and Figure 8 shows the corresponding quantitative data of the area covered by biofouling. Within 1 month, the surface of control PVC panel is totally covered by various organisms mainly consisting of tubeworms, bryozoans, and algae, indicating a high fouling pressure in the test site. In contrast, the surfaces of Y-PU-*x* are relatively clean at the early stage because of the surface renewal and release of butenolide. After immersion for 3 months, the surfaces of PLA-PU-0, PLA-PU-4, and PLA-PU-10 with TSA less than 16 wt % are almost fouled with ~95% covered area. In contrast, the surface of PLGA-PU-16 has only ~7% area covered, indicating its remarkable antifouling ability. This is because the surface of PLGA-PU-16 is renewed timely thanks to the degradation in main chain and hydrolysis in side group. The relatively fast surface renewal and the release of sufficient butenolide inhibit the fouling from adhering on the surface.

CONCLUSIONS

We have proposed a novel strategy for anti-biofouling “from the nature, for the nature” and developed an eco-friendly coating consisting of bio-derived PLA-based polyurethane and natural antifoulant butenolide. The polyurethane has good film forming property and exhibits excellent adhesion strength on epoxy substrate. By varying the kinds of degradable soft segments and the content of hydrolysable side groups, the polyurethane shows tunable hydrolytic degradation in seawater. With timely surface renewal ability, the polyurethane can act as a carrier of butenolide, which can be released in a controlled and continuous manner. Marine field test demonstrates that such an antifouling system has good antifouling ability because of the synergetic effect of surface renewal and release of butenolide. This system may find applications in high-performance and copper-free antifouling coatings.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.9b06917>.

¹H NMR, FTIR, gel permeation chromatography (GPC), and DSC spectra of Y-PU-*x*; DSC spectra of PLA diol and PLGA diols; surface morphology of Y-PU-*x* and their coatings with 10 wt % butenolide; adhesion strength of PLGA-PU-0 on epoxy substrate, Bode plots and Nyquist plots of Y-PU-*x*, time dependence of WCA of PLGA-PU-0, and time dependence of mass loss of PLGA-PU-0 in ASW at 25 °C (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The financial support of National Natural Science Foundation of China (51673074 and 51573061) is acknowledged.

REFERENCES

- (1) Lejars, M.; Margaillan, A.; Bressy, C. Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings. *Chem. Rev.* **2012**, *112*, 4347–4390.
- (2) Yebra, D. M.; Kiil, S.; Dam-Johansen, K. Antifouling technology-past, present and future steps towards efficient and environmentally friendly antifouling coatings. *Prog. Org. Coat.* **2004**, *50*, 75–104.
- (3) Schultz, M. P.; Bendick, J. A.; Holm, E. R.; Hertel, W. M. Economic impact of biofouling on a naval surface ship. *Biofouling* **2011**, *27*, 87–98.
- (4) Xie, Q.; Ma, C.; Liu, C.; Ma, J.; Zhang, G. Poly-(dimethylsiloxane)-Based Polyurethane with Chemically Attached Antifoulants for Durable Marine Antibiofouling. *ACS Appl. Mater. Interfaces* **2015**, *7*, 21030–21037.
- (5) Lejars, M.; Margaillan, A.; Bressy, C. Well-defined graft copolymers of tert-butyl dimethylsilyl methacrylate and poly-(dimethylsiloxane) macromonomers synthesized by RAFT polymerization. *Polym. Chem.* **2013**, *4*, 3282–3292.
- (6) Sommer, S.; Ekin, A.; Webster, D. C.; Staflieni, S. J.; Daniels, J.; Vanderwal, L. J.; Thompson, S. E. M.; Callow, M. E.; Callow, J. A. A preliminary study on the properties and fouling-release performance of siloxane-polyurethane coatings prepared from poly-(dimethylsiloxane) (PDMS) macromers. *Biofouling* **2010**, *26*, 961–972.
- (7) Martinelli, E.; Sarvothaman, M. K.; Alderighi, M.; Galli, G.; Mielczarski, E.; Mielczarski, J. A. PDMS network blends of amphiphilic acrylic copolymers with poly(ethylene glycol)-fluoroalkyl side chains for fouling-release coatings. I. Chemistry and stability of the film surface. *J. Polym. Sci., Part A: Polym. Chem.* **2012**, *50*, 2677–2686.
- (8) Galhenage, T. P.; Webster, D. C.; Moreira, A. M. S.; Burgett, R. J.; Staflieni, S. J.; Vanderwal, L.; Finlay, J. A.; Franco, S. C.; Clare, A. S. Poly(ethylene glycol)-modified, amphiphilic, siloxane-polyurethane coatings and their performance as fouling-release surfaces. *J. Coat. Technol. Res.* **2017**, *14*, 307–322.
- (9) Liu, S. Q.; Yang, C.; Huang, Y.; Ding, X.; Li, Y.; Fan, W. M.; Hedrick, J. L.; Yang, Y.-Y. Antimicrobial and antifouling hydrogels formed in situ from polycarbonate and poly(ethylene glycol) via Michael addition. *Adv. Mater.* **2012**, *24*, 6484–6489.
- (10) Hong, F.; Xie, L.; He, C.; Liu, J.; Zhang, G.; Wu, C. Novel hybrid anti-biofouling coatings with a self-peeling and self-generated micro-structured soft and dynamic surface. *J. Mater. Chem. B* **2013**, *1*, 2048–2055.
- (11) Magin, C. M.; Finlay, J. A.; Clay, G.; Callow, M. E.; Callow, J. A.; Brennan, A. B. Antifouling performance of cross-linked hydrogels: refinement of an attachment model. *Biomacromolecules* **2011**, *12*, 915–922.
- (12) Ekblad, T.; Bergström, G.; Ederth, T.; Conlan, S. L.; Mutton, R.; Clare, A. S.; Wang, S.; Liu, Y.; Zhao, Q.; D'Souza, F.; Donnelly, G.

- T.; Willemsen, P. R.; Pettitt, M. E.; Callow, M. E.; Callow, J. A.; Liedberg, B. Poly(ethylene glycol)-containing hydrogel surfaces for antifouling applications in marine and freshwater environments. *Biomacromolecules* **2008**, *9*, 2775–2783.
- (13) Cao, X.; Pettitt, M. E.; Wode, F.; Arpa Sancet, M. P.; Fu, J.; Ji, J.; Callow, M. E.; Callow, J. A.; Rosenhahn, A.; Grunze, M. Interaction of zoospores of the green alga *Ulva* with bioinspired micro- and nanostructured surfaces prepared by polyelectrolyte layer-by-layer self-assembly. *Adv. Funct. Mater.* **2010**, *20*, 1984–1993.
- (14) Xie, Q.; Xie, Q.; Pan, J.; Ma, C.; Zhang, G. Biodegradable polymer with hydrolysis-induced zwitterions for antibiofouling. *ACS Appl. Mater. Interfaces* **2018**, *10*, 11213–11220.
- (15) Bodkhe, R. B.; Stafslie, S. J.; Daniels, J.; Cilz, N.; Muelhberg, A. J.; Thompson, S. E. M.; Callow, M. E.; Callow, J. A.; Webster, D. C. Zwitterionic siloxane-polyurethane fouling-release coatings. *Prog. Org. Coat.* **2015**, *78*, 369–380.
- (16) Zhang, Z.; Finlay, J. A.; Wang, L.; Gao, Y.; Callow, J. A.; Callow, M. E.; Jiang, S. Polysulfobetaine-grafted surfaces as environmentally benign ultralow fouling marine coatings. *Langmuir* **2009**, *25*, 13516–13521.
- (17) Chen, S.; Jiang, S. A New Avenue to Nonfouling Materials. *Adv. Mater.* **2008**, *20*, 335–338.
- (18) Laschewsky, A.; Rosenhahn, A. Molecular design of zwitterionic polymer interfaces: searching for the difference. *Langmuir* **2019**, *35*, 1056–1071.
- (19) Quintana, R.; Gosa, M.; Jańczewski, D.; Kutnyansky, E.; Vancso, G. J. Enhanced stability of low fouling zwitterionic polymer brushes in seawater with diblock architecture. *Langmuir* **2013**, *29*, 10859–10867.
- (20) Wang, Y.; Pitet, L. M.; Finlay, J. A.; Brewer, L. H.; Cone, G.; Betts, D. E.; Callow, M. E.; Callow, J. A.; Wendt, D. E.; Hillmyer, M. A.; DeSimone, J. M. Investigation of the role of hydrophilic chain length in amphiphilic perfluoropolyether/poly(ethylene glycol) networks: towards high-performance antifouling coatings. *Biofouling* **2011**, *27*, 1139–1150.
- (21) van Zoelen, W.; Buss, H. G.; Ellebracht, N. C.; Lynd, N. A.; Fischer, D. A.; Finlay, J.; Hill, S.; Callow, M. E.; Callow, J. A.; Kramer, E. J.; Zuckermann, R. N.; Segalman, R. A. Sequence of hydrophobic and hydrophilic residues in amphiphilic polymer coatings affects surface structure and marine antifouling/fouling release properties. *ACS Macro Lett.* **2014**, *3*, 364–368.
- (22) Weinman, C. J.; Finlay, J. A.; Park, D.; Paik, M. Y.; Krishnan, S.; Sundaram, H. S.; Dimitriou, M.; Sohn, K. E.; Callow, M. E.; Callow, J. A.; Handlin, D. L.; Willis, C. L.; Kramer, E. J.; Ober, C. K. ABC triblock surface active block copolymer with grafted ethoxylated fluoroalkyl amphiphilic side chains for marine antifouling/fouling-release applications. *Langmuir* **2009**, *25*, 12266–12274.
- (23) Krishnan, S.; Weinman, C. J.; Ober, C. K. Advances in polymers for anti-biofouling surfaces. *J. Mater. Chem.* **2008**, *18*, 3405–3413.
- (24) Gudipati, C. S.; Greenlief, C. M.; Johnson, J. A.; Prayongpan, P.; Wooley, K. L. Hyperbranched fluoropolymer and linear poly(ethylene glycol) based amphiphilic crosslinked networks as efficient antifouling coatings: An insight into the surface compositions, topographies, and morphologies. *J. Polym. Sci., Part A: Polym. Chem.* **2004**, *42*, 6193–6208.
- (25) Shivapooja, P.; Wang, Q.; Orihuela, B.; Rittschof, D.; López, G. P.; Zhao, X. Bioinspired surfaces with dynamic topography for active control of biofouling. *Adv. Mater.* **2013**, *25*, 1430–1434.
- (26) Amini, S.; Kolle, S.; Petrone, L.; Ahanotu, O.; Sunny, S.; Sutanto, C. N.; Hoon, S.; Cohen, L.; Weaver, J. C.; Aizenberg, J.; Vogel, N.; Miserez, A. Preventing mussel adhesion using lubricant-infused materials. *Science* **2017**, *357*, 668–673.
- (27) Xie, Q.; Pan, J.; Ma, C.; Zhang, G. Dynamic surface antifouling: mechanism and systems. *Soft Matter* **2019**, *15*, 1087–1107.
- (28) Gevaux, L.; Lejars, M.; Margailan, A.; Bressy, C. Water erodible coatings based on a hydrolyzable PDMS/polyester network. *Mater. Today Commun.* **2018**, *17*, 517–526.
- (29) Sheng, X.; Wu, W.; Qin, Y.; Wang, X.; Wang, F. Efficient synthesis and stabilization of poly(propylene carbonate) from delicately designed bifunctional aluminum porphyrin complexes. *Polym. Chem.* **2015**, *6*, 4719–4724.
- (30) Wang, Y.; Qin, Y.; Wang, X.; Wang, F. Trivalent titanium salen complex: thermally robust and highly active catalyst for copolymerization of CO₂ and cyclohexene oxide. *ACS Catal.* **2015**, *5*, 393–396.
- (31) Pang, X.; Zhuang, X.; Tang, Z.; Chen, X. Polylactic acid (PLA): research, development and industrialization. *Biotechnol. J.* **2010**, *5*, 1125–1136.
- (32) Lasprilla, A. J. R.; Martinez, G. A. R.; Lunelli, B. H.; Jardini, A. L.; Filho, R. M. Poly-lactic acid synthesis for application in biomedical devices - A review. *Biotechnol. Adv.* **2012**, *30*, 321–328.
- (33) Santoro, M.; Shah, S. R.; Walker, J. L.; Mikos, A. G. Poly(lactic acid) nanofibrous scaffolds for tissue engineering. *Adv. Drug Delivery Rev.* **2016**, *107*, 206–212.
- (34) Hu, H.; Zhang, R.; Shi, L.; Ying, W. B.; Wang, J.; Zhu, J. Modification of poly(butylene 2,5-furandicarboxylate) with lactic acid for biodegradable copolyesters with good mechanical and barrier properties. *Ind. Eng. Chem. Res.* **2018**, *57*, 11020–11030.
- (35) Zhang, L.; Dong, H.; Li, M.; Wang, L.; Liu, Y.; Wang, L.; Fu, S. Fabrication of polylactic acid-modified carbon black composites into improvement of levelness and mechanical properties of spun-dyeing polylactic acid composites membrane. *ACS Sustainable Chem. Eng.* **2019**, *7*, 688–696.
- (36) Martin, R. T.; Camargo, L. P.; Miller, S. A. Marine-degradable polylactic acid. *Green Chem.* **2014**, *16*, 1768–1773.
- (37) Li, C.-Y.; Chen, J.-H.; Chien, P.-C.; Chiu, W.-Y.; Chen, R.-S.; Don, T.-M. Preparation of poly(IPDI-PTMO-Siloxanes) and influence of siloxane structure on reactivity and mechanical properties. *Polym. Eng. Sci.* **2007**, *47*, 625–632.
- (38) Gentile, P.; Chiono, V.; Carmagnola, I.; Hatton, P. An overview of poly(lactic-co-glycolic) acid (PLGA)-based biomaterials for bone tissue engineering. *Int. J. Mol. Sci.* **2014**, *15*, 3640–3659.
- (39) Ma, C.; Zhang, W.; Zhang, G.; Qian, P.-Y. Environmentally friendly antifouling coatings based on biodegradable polymer and natural antifoulant. *ACS Sustainable Chem. Eng.* **2017**, *5*, 6304–6309.
- (40) Ding, W.; Ma, C.; Zhang, W.; Chiang, H.; Tam, C.; Xu, Y.; Zhang, G.; Qian, P.-Y. Anti-biofilm effect of a butenolide/polymer coating and metatranscriptomic analyses. *Biofouling* **2018**, *34*, 111–122.
- (41) Dahms, H.; Dobretsov, S. Antifouling compounds from marine macroalgae. *Mar. Drugs* **2017**, *15*, 265–280.
- (42) Feng, D. Q.; He, J.; Chen, S. Y.; Su, P.; Ke, C. H.; Wang, W. The plant alkaloid camptothecin as a novel antifouling compound for marine paints: laboratory bioassays and field trials. *Mar. Biotechnol.* **2018**, *20*, 623–638.
- (43) Lau, S. C. K.; Qian, P. Y. Inhibitory effect of phenolic compounds and marine bacteria on larval settlement of the barnacle *Balanus amphitrite* darwin. *Biofouling* **2000**, *16*, 47–58.
- (44) Bellotti, N.; Deyá, C.; del Amo, B.; Romagnoli, R. Antifouling Paints with Zinc "Tannate". *Ind. Eng. Chem. Res.* **2010**, *49*, 3386–3390.
- (45) Norcy, T. L.; Niemann, H.; Proksch, P.; Linossier, I.; Vallée-Réhel, K.; Hellio, C.; Fay, F. Anti-biofilm effect of biodegradable coatings based on hemibastadin derivative in marine environment. *Int. J. Mol. Sci.* **2017**, *18*, 1520–1539.
- (46) Xu, Y.; He, H.; Schulz, S.; Liu, X.; Fusetani, N.; Xiong, H.; Xiao, X.; Qian, P.-Y. Potent antifouling compounds produced by marine *Streptomyces*. *Bioresour. Technol.* **2010**, *101*, 1331–1336.
- (47) Zhang, Y.-F.; Xiao, K.; Chandramouli, K. H.; Xu, Y.; Pan, K.; Wang, W.-X.; Qian, P.-Y. Acute toxicity of the antifouling compound butenolide in non-target organisms. *PLoS One* **2011**, *6*, No. e23803.
- (48) Ma, C.; Xu, W.; Pan, J.; Xie, Q.; Zhang, G. Degradable polymers for marine antibiofouling: optimizing structure to improve performance. *Ind. Eng. Chem. Res.* **2016**, *55*, 11495–11501.
- (49) Regina, V. R.; Søhoel, H.; Lokanathan, A. R.; Bischoff, C.; Kingshott, P.; Revsbech, N. P.; Meyer, R. L. Entrapment of subtilisin

in ceramic sol-gel coating for antifouling applications. *ACS Appl. Mater. Interfaces* **2012**, *4*, 5915–5921.

(50) Wang, G.; Wang, L.; Lin, W.; Wang, Z.; Zhang, J.; Ji, F.; Ma, G.; Yuan, Z.; Chen, S. Development of robust and recoverable ultralow-fouling coatings based on poly(carboxybetaine) ester analogue. *ACS Appl. Mater. Interfaces* **2015**, *7*, 16938–16945.

(51) Qi, M.; Song, Q.; Zhao, J.; Ma, C.; Zhang, G.; Gong, X. Three-dimensional bacterial behavior near dynamic surfaces formed by degradable polymers. *Langmuir* **2017**, *33*, 13098–13104.

(52) Qi, M.; Gong, X.; Wu, B.; Zhang, G. Landing dynamics of swimming bacteria on a polymeric surface: effect of surface properties. *Langmuir* **2017**, *33*, 3525–3533.

(53) Bellotti, N.; Romagnoli, R. Assessment of zinc salicylate as antifouling product for marine coatings. *Ind. Eng. Chem. Res.* **2014**, *53*, 14559–14564.

(54) Tribou, M.; Swain, G. Grooming using rotating brushes as a proactive method to control ship hull fouling. *Biofouling* **2015**, *31*, 309–319.

(55) Chen, S.; Ma, C.; Zhang, G. Biodegradable polymer as controlled release system of organic antifoulant to prevent marine biofouling. *Prog. Org. Coat.* **2017**, *104*, 58–63.

(56) Dai, G.; Xie, Q.; Ma, C.; Zhang, G. Biodegradable poly(ester-co-acrylate) with antifoulant pendant groups for marine anti-biofouling. *ACS Appl. Mater. Interfaces* **2019**, *11*, 11947–11953.

(57) Xu, W.; Ma, C.; Ma, J.; Gan, T.; Zhang, G. Marine biofouling resistance of polyurethane with biodegradation and hydrolyzation. *ACS Appl. Mater. Interfaces* **2014**, *6*, 4017–4024.

(58) Badami, A. S.; Kreke, M. R.; Thompson, M. S.; Riffle, J. S.; Goldstein, A. S. Effect of fiber diameter on spreading, proliferation, and differentiation of osteoblastic cells on electrospun poly(lactic acid) substrates. *Biomaterials* **2006**, *27*, 596–606.

(59) Xie, Q.; Ma, C.; Zhang, G.; Bressy, C. Poly(ester)-poly(silyl methacrylate) copolymers: synthesis and hydrolytic degradation kinetics. *Polym. Chem.* **2018**, *9*, 1448–1454.

(60) Bressy, C.; NGuyen, M. N.; Tanguy, B.; Ngo, V. G.; Margaillan, A. Poly(trialkylsilyl methacrylate)s: a family of hydrolysable polymers with tuneable erosion profiles. *Polym. Degrad. Stab.* **2010**, *95*, 1260–1268.

(61) Catiker, E.; Gümüşderelioğlu, M.; Güner, A. Degradation of PLA, PLGA homo- and copolymers in the presence of serum albumin: a spectroscopic investigation. *Polym. Int.* **2000**, *49*, 728–734.

(62) Reed, A. M.; Gilding, D. K. Biodegradable polymers for use in surgery - poly(glycolic)/poly(lactic acid) homo and copolymers: 2. In vitro degradation. *Polymer* **1981**, *22*, 494–498.